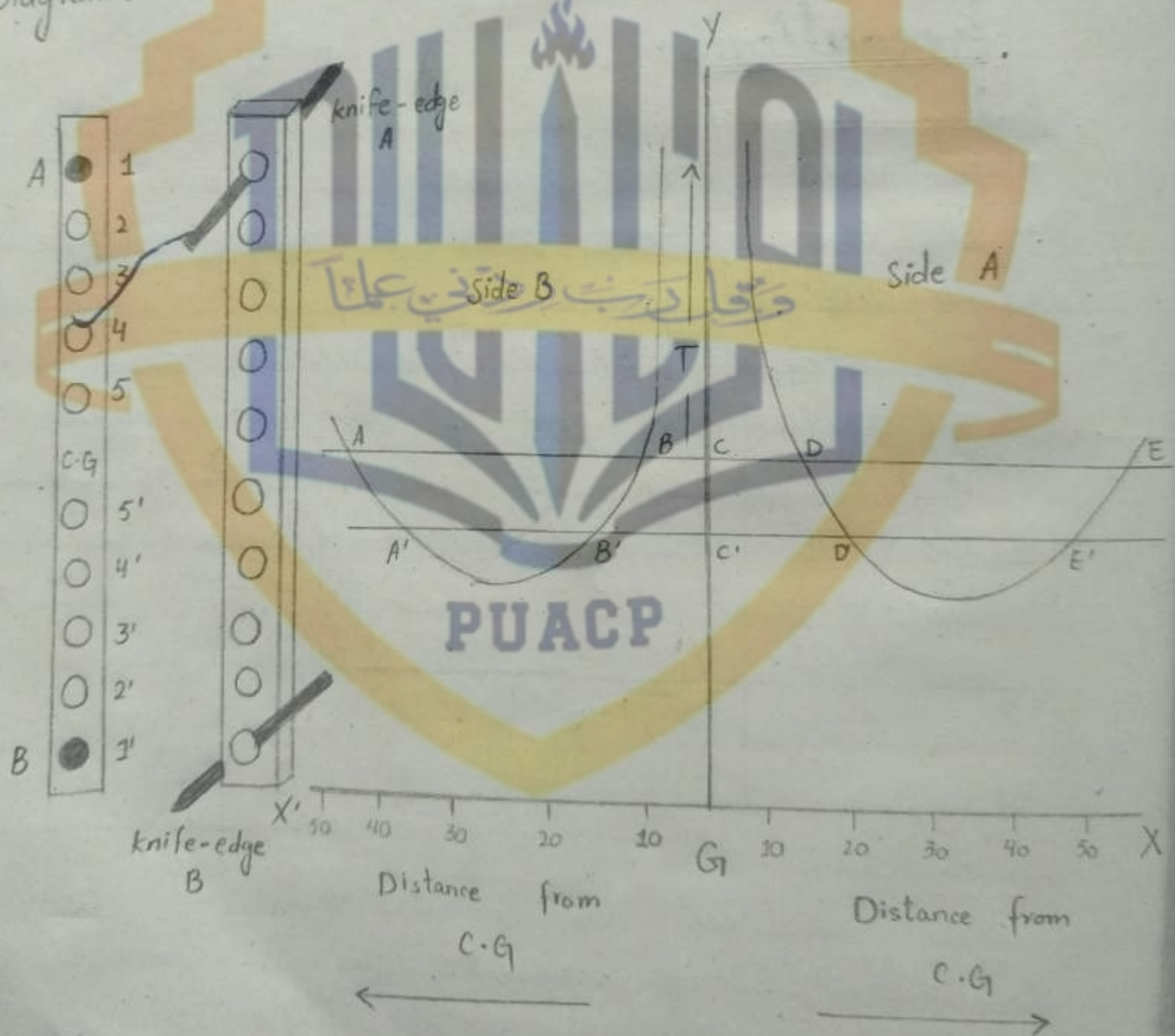


Experiment : Determine the value of g by means of a Compound pendulum

Apparatus : • Compound Pendulum with two knife edges
• Support • Beam Compass • stop watch • Sharp edges
• Telescope • Spirit level • meter rod

Diagram :



Experiment Name: **Determine the value of g by means of a Compound Pendulum**

Apparatus :-

- Compound Pendulum with two knife edges
- Support
- Stop watch
- Beam compass
- Telescope
- Meter rod
- Sharp edges
- Spirit level

Procedure :-

- Mark the knife edges of the bar of pendulum as A and B with the piece of chalk.
- Balance the Pendulum on the wedge, with one - knife - edge at extreme hole on one side and other knife - edge at extreme hole on the other side mark the position of its centre of gravity.
- level the stand in horizontal position with the help of spirit level. and suspend the pendulum on it from the knife edges A. Draw cross at Both end of the pendulum or draw a vertical line.
- With the help of your hand, adjust eye - piece of telescope till the cross wire are seen correctly.

Observation and Calculation

For Edge A

Hole no.	Time for 20 vibration			Time Period $T = t/20$ sec	Distance from C.G. in cm
	1 sec	2 sec	Mean (t) sec		
1.	31.5	31.3	31.4	1.57	45
2.	31.2	31.3	31.2	1.56	40
3.	30.6	30.8	30.7	1.50	35
4.	30.4	30.2	30.3	1.51	30
5.	29.3	29.4	29.3	1.46	25
6.	30.5	30.4	30.9	1.54	20
7.	32.4	32.7	32.5	1.62	15
8.	37.3	37.4	37.3	1.86	10
9.	48.2	48.3	48.2	2.4	5

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For Edge B

Hole no.	Time For 20 vibration			Time Period $T = t/20$ sec	Distance from C.G. in cm
	1 sec	2 sec	Mean (t) sec		
1.	31	31	31.1	1.55	45
2.	30.6	30.7	30.6	1.53	40
3.	31.9	31.8	31.8	1.52	35
4.	30.1	30.2	30.1	1.51	30
5.	30.7	30.8	30.7	1.52	25
6.	30.8	30.9	30.8	1.54	20
7.	33.1	33.2	33.1	1.65	15
8.	37.5	37.6	37.5	1.87	10
9.	50.1	49.8	49.9	2.4	5

- Now Focus the telescope across the reference line of pendulum from distance of about 3 meter's.
- Displace the pendulum on one-side through small angle of about (5°) and release it.
- looking through the telescope find time for suitable no. of vibrations by
- stop watch.
- Measure the distance between the inner edge of knife - edge A and centre of gravity G.
- Now shift the upper knife - edge A to the second hole and shift the lower edge B to the second hole from below to keep the C.G in its initial position.
- Repeat this process to note the time of same Number of vibration.
- Now turn the pendulum ups and down and take a set of observations on the other half of length of pendulum.

Calculation of 'g' from Graph

No. of Obs.	Length of equivalent simple pendulum from Graph L (cm)	Time Period T (sec)	$g = 4\pi^2 \frac{L}{T^2}$ cm - sec ⁻²
1	$L = \frac{1}{2}(AD + BE)$ $AD = AB + CD = 47 + 13 = 60$ $BE = BC + DE = 15 + 44 = 59$ $L = \frac{1}{2}(60 + 59)$ $L = 59.5$	$T = GC = 1.7$	$g = \frac{4(3.14)^2(59.5)}{(1.7)^2}$ $g = \frac{2348.96}{2.89}$ $g = 812.78$
2	$L = \frac{1}{2}(A'D' + B'E')$ $A'D' = A'B' + C'D' = 40 + 20 = 60$ $B'E' = B'C' + D'E' = 22 + 37 = 59$ $L = \frac{1}{2}(60 + 59)$ $L = 59.5$	$T = G'C' = 1.5$	$g = \frac{4(3.14)^2(59.5)}{(1.5)^2}$ $g = \frac{2348.96}{2.25}$ $g = 1043.98$

Mean Value of $g = \frac{812.78 + 1043.98}{2} = 928.38 \text{ cm-sec}^{-2}$

Result :

Mean Value of $g = 928.39 \text{ cm-sec}^{-2} = \frac{928.39}{100} = 9.28 \text{ ms}^{-2}$

Actual Value of $g = 9.8 \text{ ms}^{-2}$

Percentage error = $\frac{9.8 - 9.3}{9.8} \times 100$
 $= 0.05 \times 100$

Percentage error = 5%

Plotting of Graphs :-

Taking Time period (T) along y-axis and distance from centre of gravity (G) along x-axis.

- Draw any line $ABCDE$ parallel to x-axis so as to cut the two curves at these points A, B, C, D and E .
- Find the length of line AB, BC, CD , and DE .
The length of simple pendulum from the graph is;

$$L_1 = BC + CE = BE = 59 \text{ cm}$$

- Also the length of equivalent simple pendulum is;

$$L_2 = DC + AG = AD = 60 \text{ cm}$$

$$\text{Mean length} = L = L_1 + L_2 = 119 \text{ cm}$$

$$L = \frac{1}{2} (BE + AD) = \frac{1}{2} (119) = 59.5$$

- Now Draw another line parallel to x-axis and cutting the graph at points A', B', C', D' and E' similarly take same process. the find value of g using;

$$g = \frac{4\pi^2 L}{T^2}$$